

# Anushka Verma

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [GeeksforGeeks](#) | [Website](#)



## Professional Summary

Final-year student with a strong foundation in Data Structures & Algorithms, Full Stack Development (PERN), and Artificial Intelligence / Machine Learning. Currently exploring the domains of System design and Cloud. Skilled in building scalable web applications and solving complex problems, with a focus on developing impactful solutions.

## Education

### Jaypee Institute of Information Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.42

Noida, Uttar Pradesh

2023-2027

### Lovely Public Senior Secondary School

Class XII, PCM + Computer Science – 90.4%

East Delhi

2022

## Experience

### NimbleBiz.ai | Software & AI Engineer Intern

Remote | Aug 2026 – Present

- Architected and developed a high-performance filtering and dynamic table customization engine serving 1,000+ users using Next.js and TypeScript.
- Programmed dynamic frontend components featuring multi-dimensional querying, drag-and-drop column reordering, and stateful views.
- Wrote and deployed production-level codebase modules for automated Voice and Chat AI conversational bots.

### YesCity.in | Software Development Intern

Remote | May 2026 – Jul 2026

- Developed an API-based discount feature for premium subscriptions using coupons integrated with Razorpay, enabling dynamic pricing.
- Automated coupon validation and discount calculation logic based on configurable business rules, reducing manual pricing errors.
- Built an automated data ingestion pipeline (web scraping and third-party APIs) that fed the research team's itinerary CRM engine, improving data-structuring efficiency.

## Projects

### Graph Community Detection Engine | [Present] C++17, Louvain, Leiden, DSU, Topological Sort, NetworKit, Python, CI/CD | NMI: 0.76 [GitHub](#)

- Engineered a Compressed Sparse Row (CSR) graph engine in C++17 and implemented the Louvain and Leiden community-detection algorithms from scratch, validated via modularity and NMI against ground-truth LFR benchmarks.
- Built a DFS-verified connectivity guarantee, a Union-Find (DSU) based query layer, and a topological-sort-validated community hierarchy for fast multi-resolution lookups.
- Benchmarked against NetworKit on a real-world SNAP dataset using a custom ID-remapping pipeline, improving NMI from 0.57 to 0.76 by tuning the modularity resolution parameter to correct for its resolution-limit bias

### Chest X-Ray Disease Detection | FastAPI, Data Preprocessing, Deep Learning, CNN, NLP, RAG, Gen AI, Agentic AI, Docker | AUC: 0.75 [GitHub](#)

- Fine-tuned DenseNet-121 DL model on the NIH ChestX-ray14 dataset for multi-label classification across 14 chest diseases.
- Integrated a RAG pipeline with a Groq LLM to auto-generate explainable diagnostic summaries from model predictions.
- Engineered Grad-CAM visualizations for model interpretability and designed an agentic layer to orchestrate inference, retrieval, and report generation; containerized the full pipeline with Docker.

### Intelligent Incident Detection and Prediction System | Flask, EDA, Machine/Deep Learning, CrewAI, Ollama | Precision: 0.99 [GitHub](#)

- Built a real-time incident detection and prediction pipeline using Random Forest and Transformer-based models.
- Designed a multi-agent workflow with CrewAI integrated with Ollama to generate root-cause analysis and incident explanations.
- Reduced manual monitoring effort by 70–80% through proactive, automated incident prediction and reporting.

## Technical Skills

**Languages:** Python, C++, C, Java, JavaScript

**Web Development:** HTML, CSS, Node.js, Express.js, React.js, EJS, jQuery, Bootstrap, FastAPI, REST API

**AI/ML & Agentic Systems:** ML/DL models, CNNs, NLP, Generative AI, RAG pipelines, Agentic AI, LangChain, OpenCV

**Databases:** PostgreSQL, MySQL

**Systems & DevOps:** Docker, Linux, AWS SSH, AWK, Git, GitHub, Bash, Shell scripting

**Core CS:** Data Structures & Algorithms, OOPs using java, Operating Systems, Computer Networks, Software Engineering

## Achievements & Certifications

- Solved 600+ problems across LeetCode, GeeksforGeeks and CodeChef.
- Senior Advisor, IIIT Knuth Programming Hub (KPH) Technical Core Team.
- Semi-finalist, Code4Cause 2.0 Hackathon, NSUT Delhi (2024).
- Full Stack Web Development (Angela Yu, Udemy) and Introduction to Python (edX); currently pursuing AI/ML (Krish Naik).